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SES598 Space Robotics and AI
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Cart Pole Control Using LQR

Introduction: The cart-pole system is a very famous example and benchmark in control theory that demonstrates the stabilization of an inherently unstable system. The objective of this assignment is to design and analyze a Linear Quadratic Regulator (LQR) controller capable of balancing a pendulum in the upright position while maintaining the carts position near the center of the 5 meter long track. The system is subjected to disturbances caused by simulated earthquakes. This means that the controller has to maintain stability under these conditions while minimizing control effort and state deviation.

Parameter Analysis: The matrices for an LQR controller that will be iterated on are the Q matrix (diagonal entries) and the R matrix (scalar value). Both of these matrices are used in the quadratic cost function for our controller.

Our goal is to minimize a quadratic cost functional:

$$J(U, x_0) = \sum_{\tau=0}^{N-1} \left(x_\tau^T Q x_\tau + u_\tau^T R u_\tau \right) + x_N^T Q_f x_N,$$

where $Q, R, Q_f \succeq 0$ are positive semidefinite matrices and $U := (u_0, u_1, \ldots, u_{N-1})$. In equation (2):

- N is called time horizon (we will consider $N = \infty$ later);
- $x_t^T Q x_t$ measures state deviation cost at $\tau = t$;
- $u_t^T R u_t$ measures input authority cost at $\tau = t$;
- $x_N^T Q_f x_N$ measures the terminal cost.

The Q matrix contains diagonal entries [x, x_dot, theta, theta_dot]. These are the parameters that we can control and measure with our cart. The Q matrix penalizes state deviations as all of these values are assumed to be zero when steady state is achieved. X is the location of the cart and X_dot is the velocity of the cart. Theta is the angle difference between the pole in the upright position and our theta_dot is the angular velocity of the pole's angle change.

Tweaking the values of the Q matrix changes how much importance we put into these parameters for the cost function. The heuristics that I used when tuning these Q parameters for the cartpole was as follows.

- increase X keeps cart near center but at cost of stability
- increase Xdot Damping to keep cart from thrashing
- increase Theta fights falling over more
- increase Thetadot adds damping to reduce oscillation

These proved valuable. However there are certain caveats that come with all heuristics, as you cannot simply live through rules-of-thumb. This will be covered later when analyzing these trials.

The R matrix penalizes control effort for our controller. The R matrix can be thought of how much total effort the controller can use to react to the sudden changes induced by the earthquake simulator. In a satellite this R matrix could be higher if we are controlling the satellite's position with thrusters, because these thrusters would have limited fuel. The heuristics for tuning the R matrix for this controller was as follows.

- Increasing R will result in a less aggressive controller
- Decreasing R will result in a more aggressive controller

System model and Design:

The state-space representation of the system was as follows $X = [X, \dot{X}, \Theta, \dot{\theta}]^T$
The control input is the horizontal force applied to the cart. The system was linearized around the upright equilibrium point and controlled using a discrete-time LQR controller.

The Feedback Law is:
$u = -Kx$
Where K is computed from the discrete algebraic Riccati equation using the weighting matrices Q and R previously defined between trial iterations.

Testing and Quality Metrics:

In order to properly test these input Q and R matrices, I wanted to see the total duration, maximum cart displacement, maximum pole angle deviation from zero, average control effort in newtons, the Root Mean Square Error of the cart position from the middle of the track, and Finally the Stability score. All of these metrics excluding the RMS of the cart position were built into the lqr_controller python file. Additionally at the end of each trial the plots are also used in debugging the parameter values for the Q and R matrices. The plots that were shown at the end of each trial were cart position, pole deviation, earthquake force (disturbance), and control input. These gave insight that would be very useful in how well the controller did at counteracting the earthquake force. However, because the earthquake force was created

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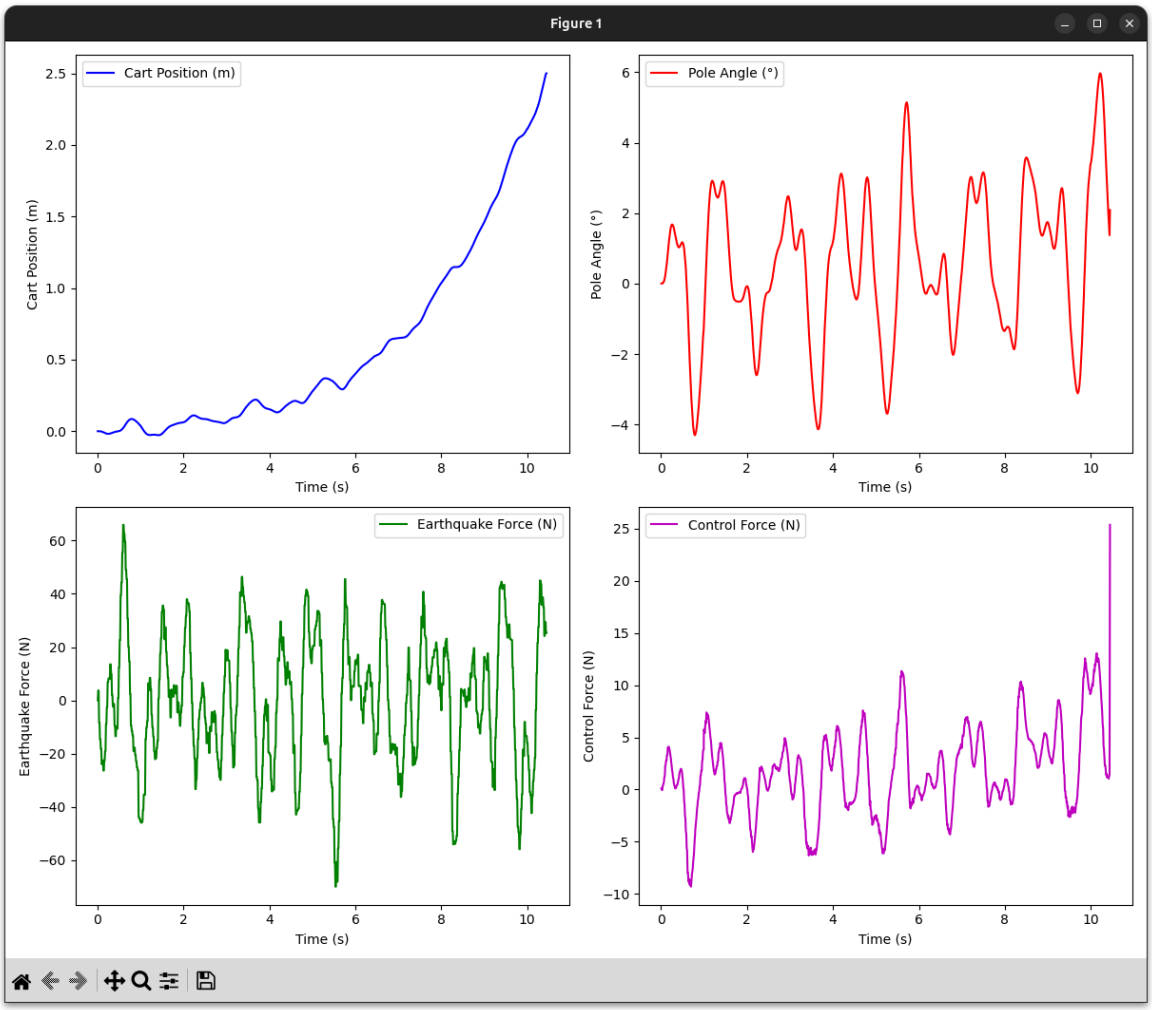

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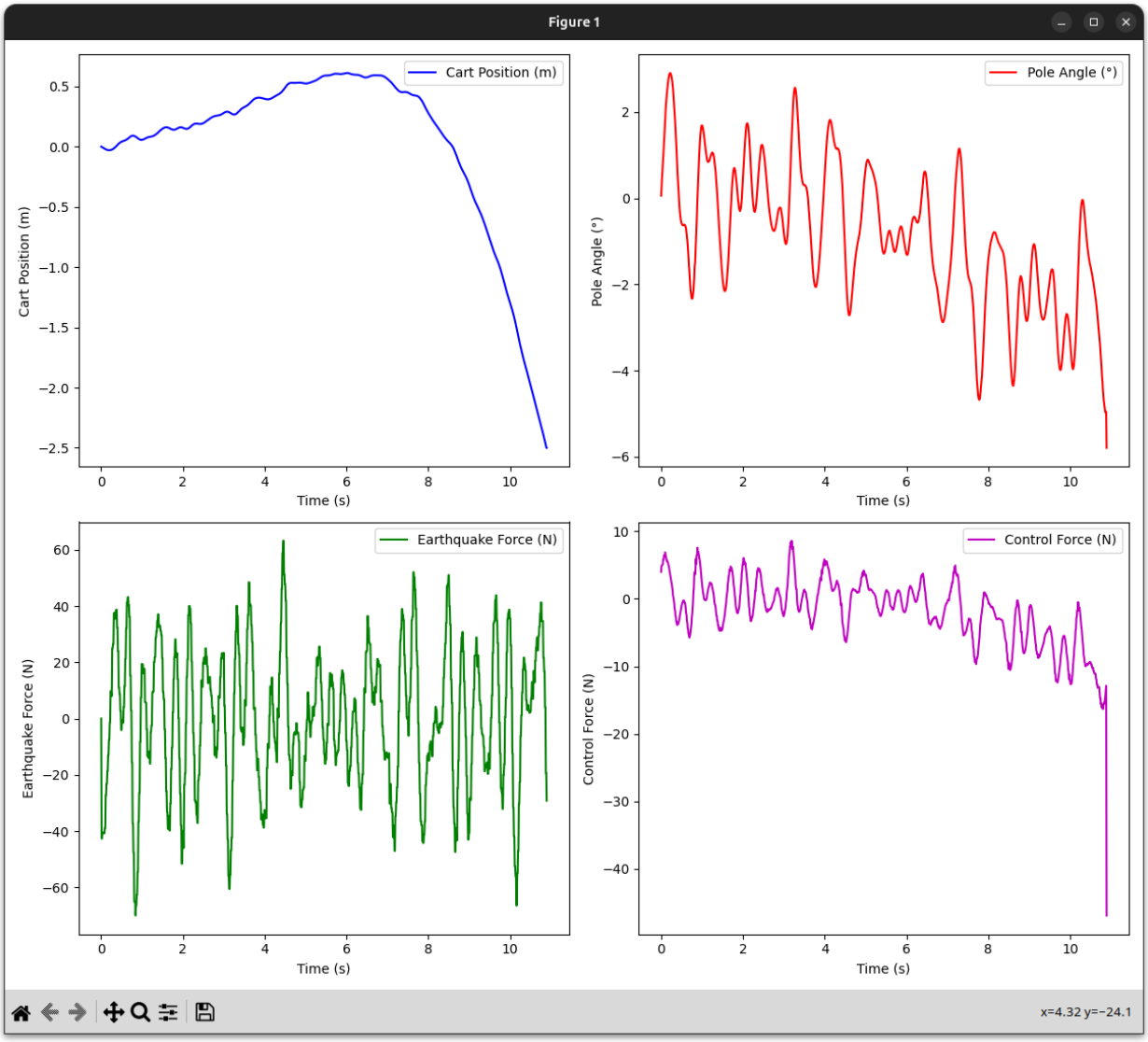

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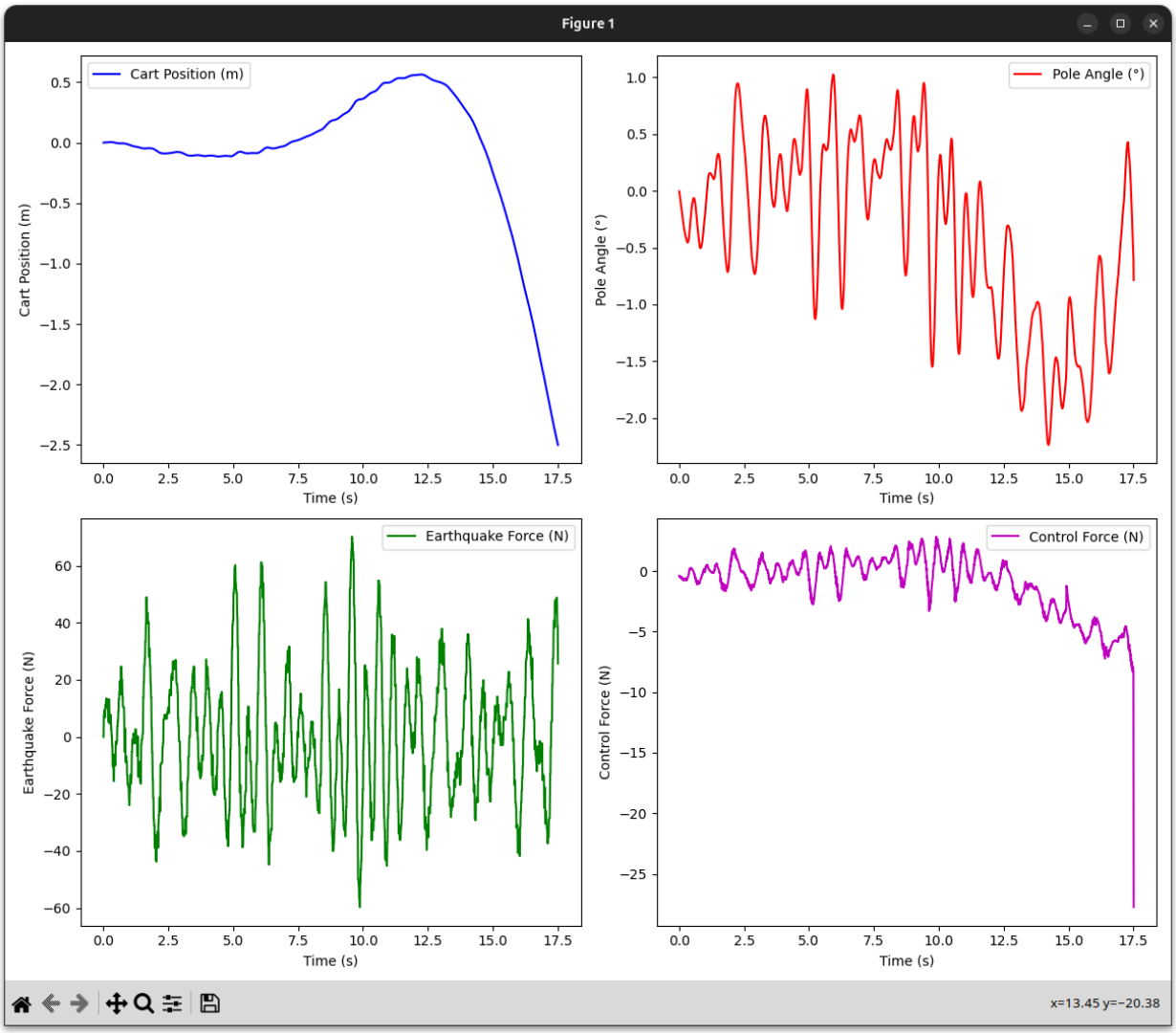

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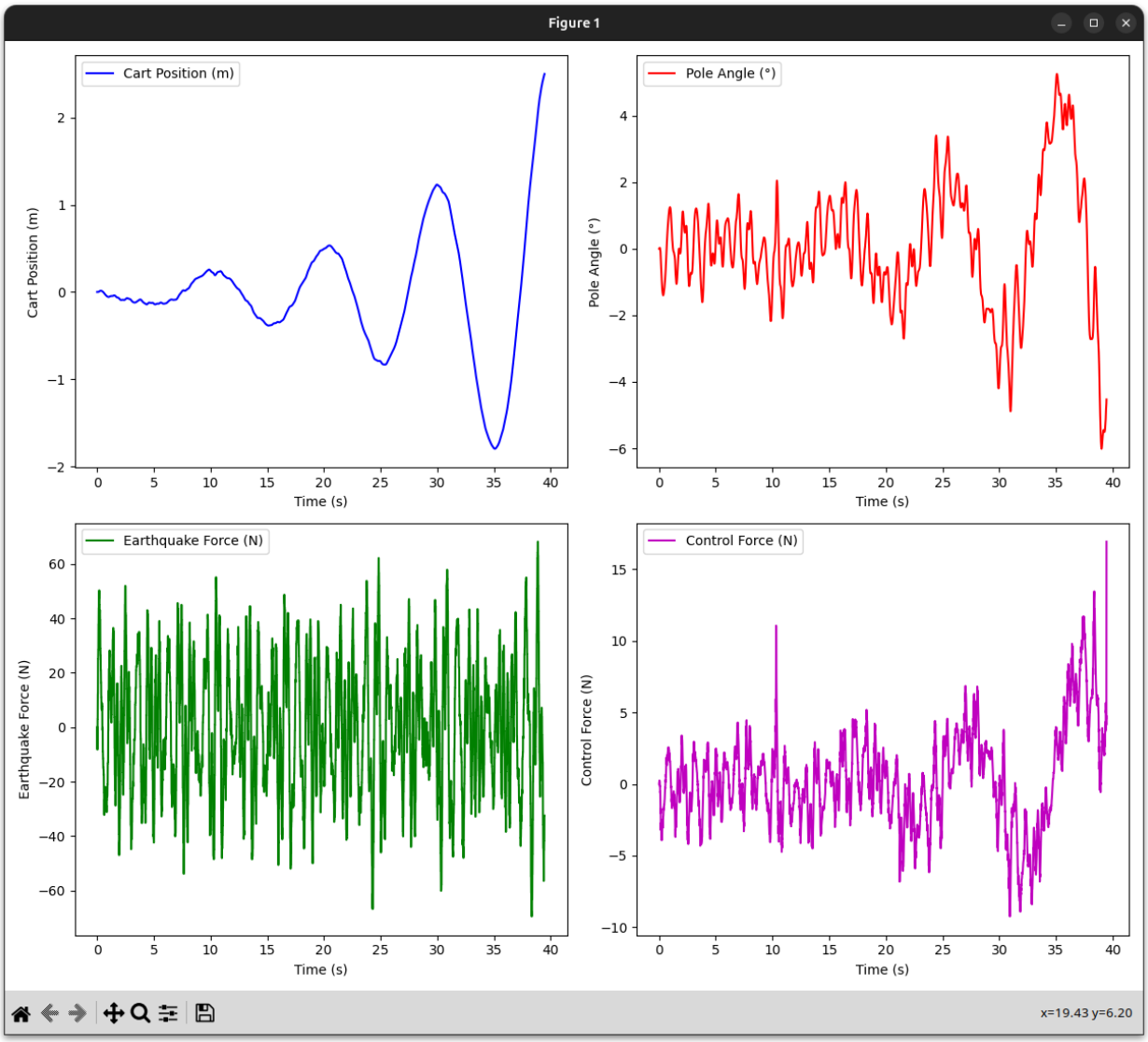

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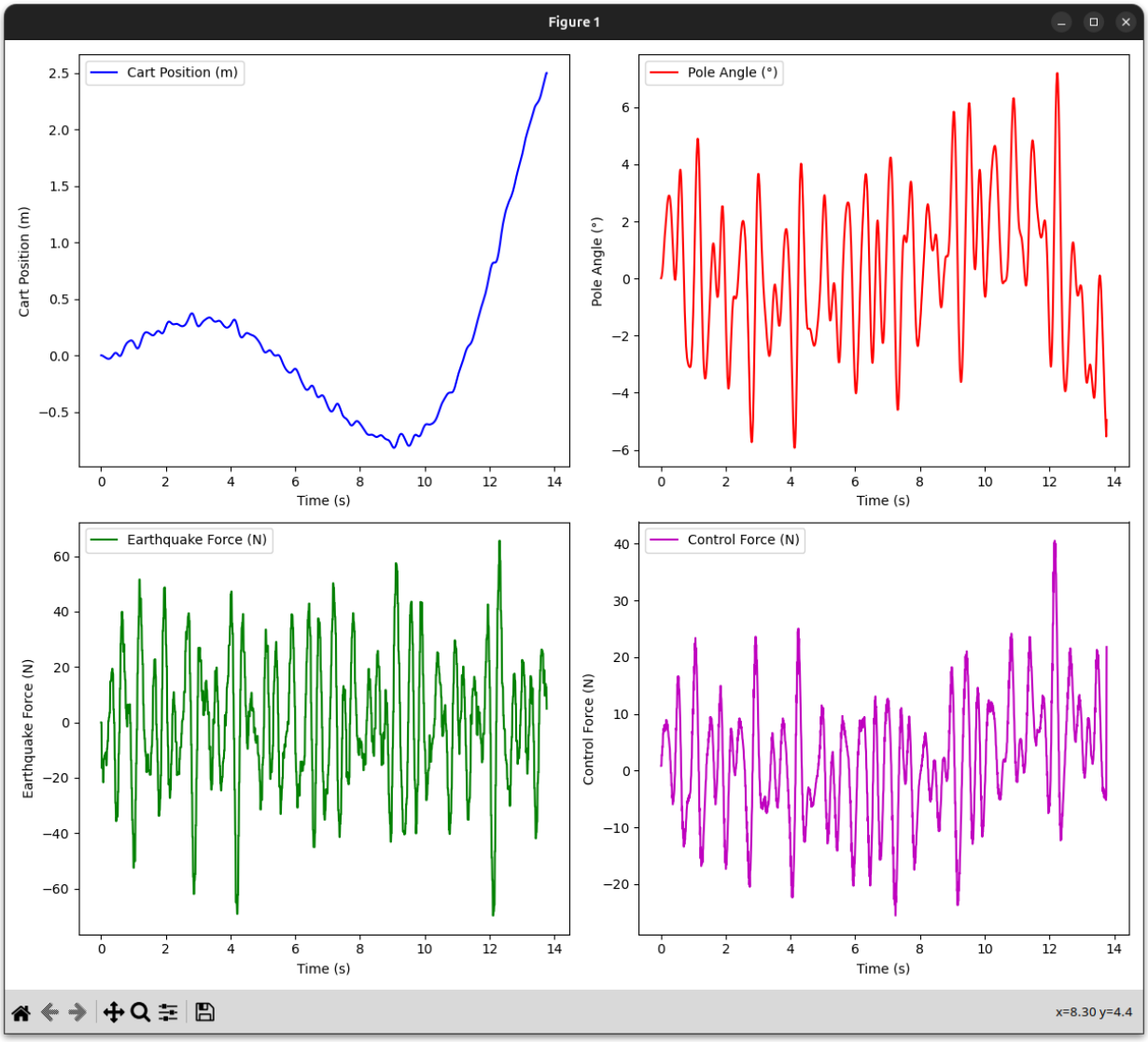

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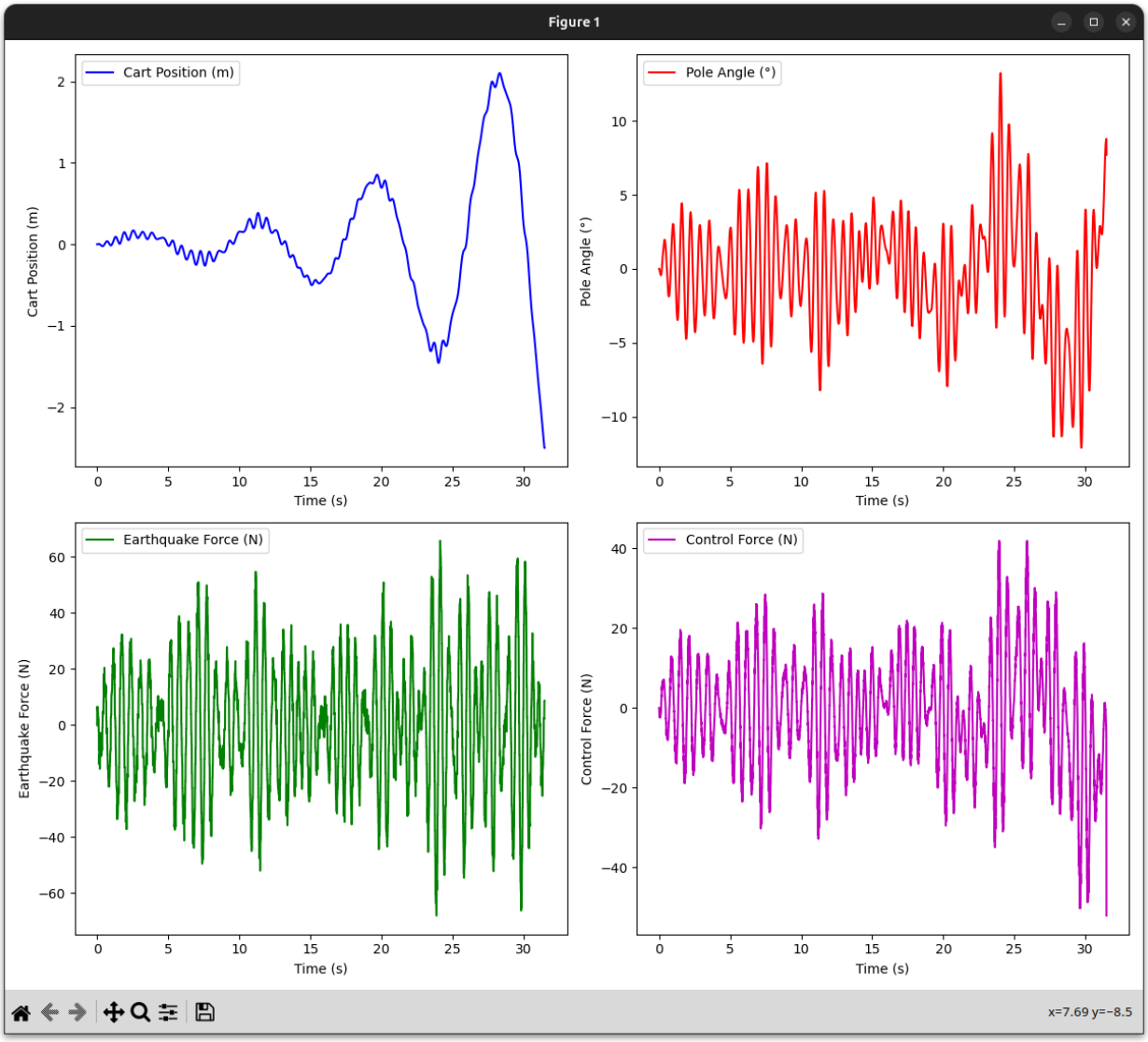

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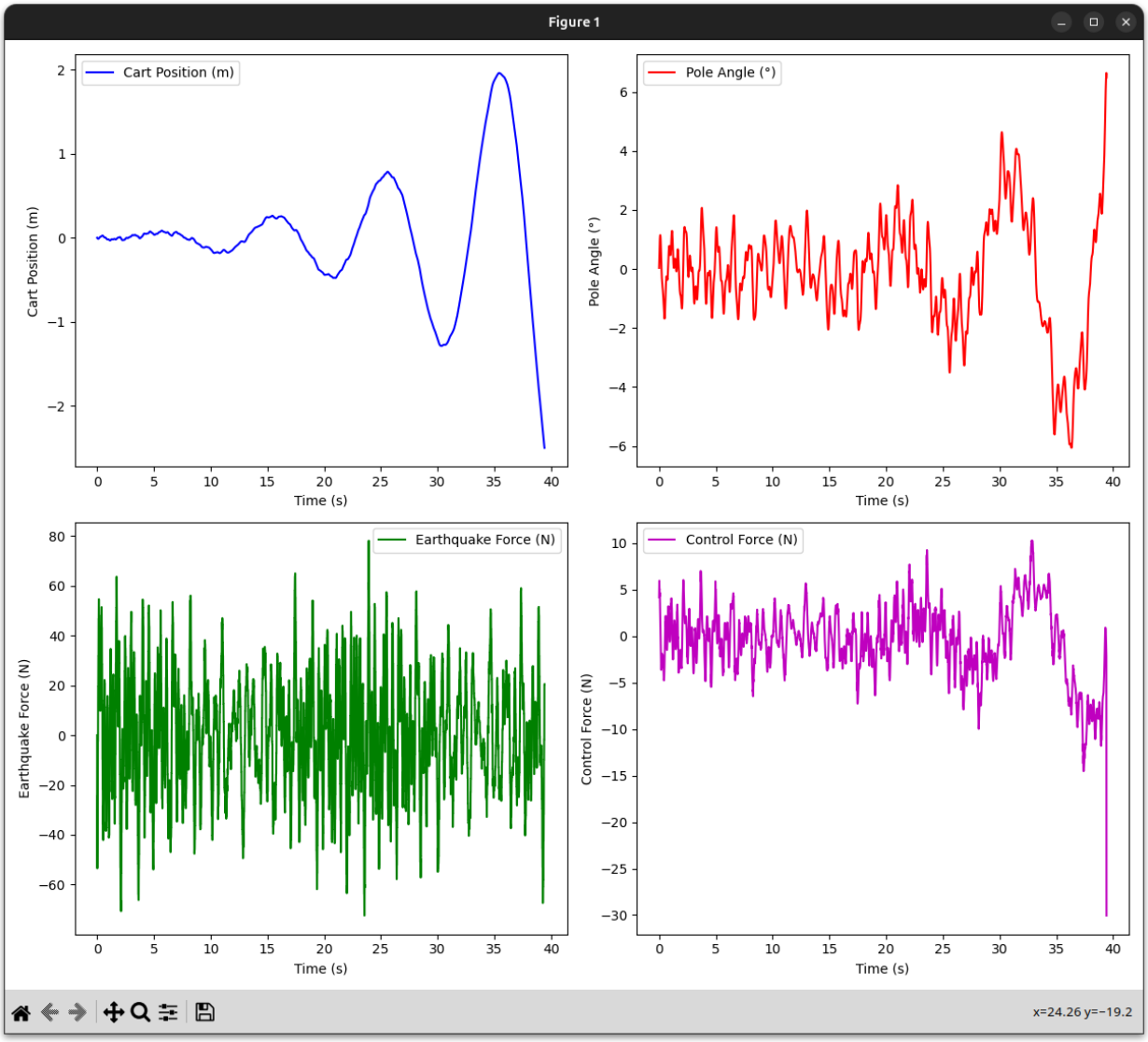

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ion=120.00s
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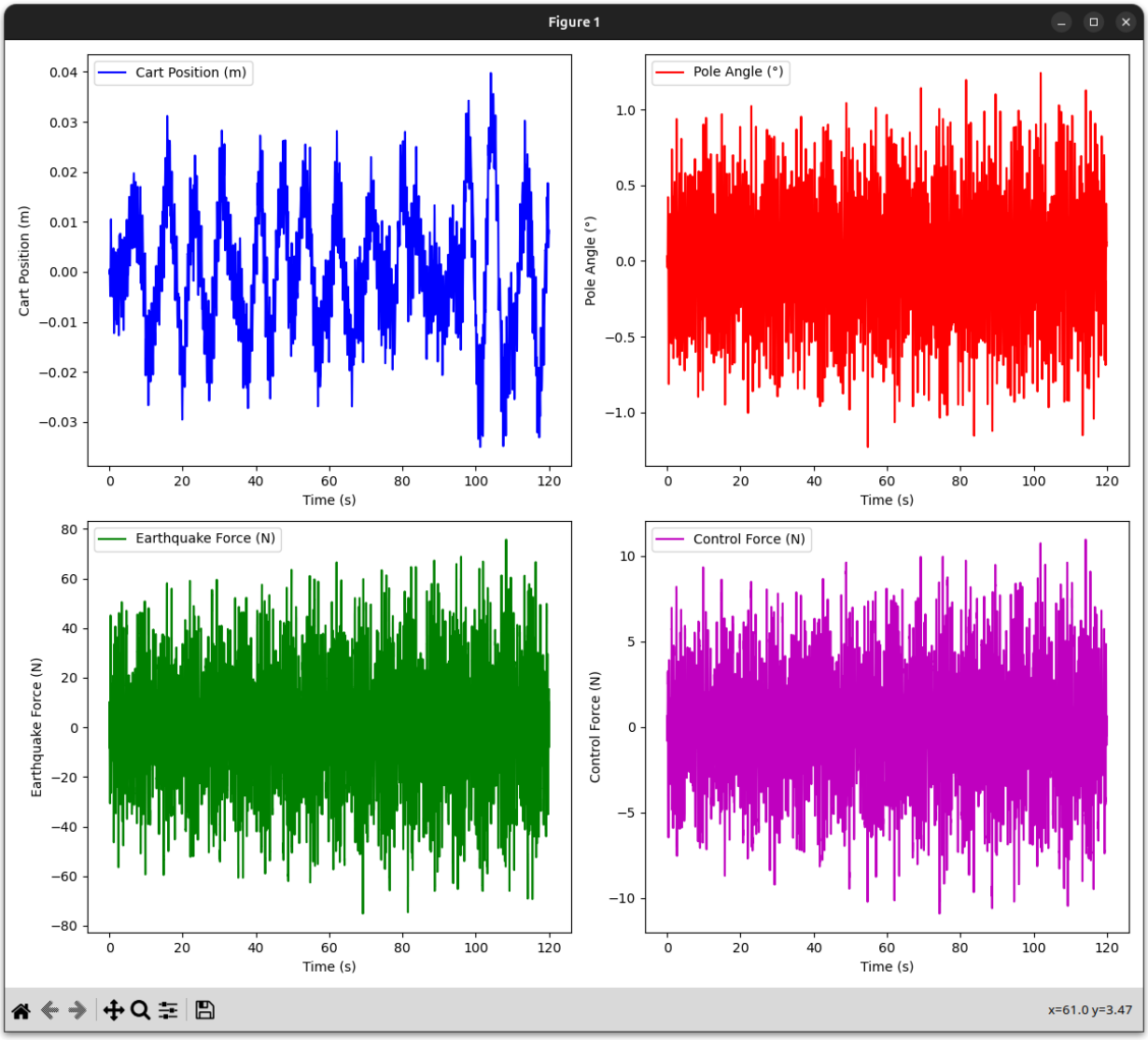

Overview of Results and conclusions:
During early trials and iterations the cart was not penalized enough for using the entirety of the track to obtain stability. The state location of the cart was drifting over time even though the beginning angle deviations were acceptable given the charts. So this meant that the penalty for the cart location needed to increase and or the R matrix that controlled aggressiveness needed to be increased to reject sudden changes from the earthquake forces.

During the middle of the trial tests I was able to create a longer duration of stability, but the Q matrix for the cart location and velocity still needed to be increased to reduce the root mean square error of the cart location as overtime the location of the cart would still drift off to the ends of the rails.

The final trial and best result lasted the full 2 minutes for the trial. The maximum displacement for the cart was roughly 0.04 meters from zero, this is really good as it meant that the cart only needed to traverse a couple inches to withstand the earthquake forces. The max pole deviation was within 1.24 degrees with an average control effort of 2.69 newtons. The root mean squared error for the pole was only 0.013 meters. This trial was the best with a stability score of 9.54/10.

The results of the final trial tell us that the main failure mode was a constraint failure coming from when the cart hit ∓ 2.5 meters. This would result in an immediate failure as the pole would topple over when it reached the end of the track. This best trial proves that the controller can achieve high disturbance rejection by finding the right Q/R tradeoff based on the heuristics of the controller described in the introduction and parameter analysis section of this report.

Future Work:
These trials were manually tuning the Q and R matrices for this cart-pole problem. However it took me roughly 3 hours to find numbers that I deemed acceptable. After some research I discovered Bryson’s Rule for systematic LQR weight selection. This approach is more attractive as it produces weights with clear physical interpretation each time. Each term in the cost function contributes similarly when the corresponding state or input reaches its acceptable limit. In my implementation I had trouble keeping the cart in the middle of the track without clear physical limits to the cart position. By using Bryson's Rule you can have the maximum limit of the cart location be 2.5 meters away from the center. This is achieved by normalizing each state

and input according to the acceptable maximum deviation.

1.5 Additional notes

Sidebar 1 (Bryson's rule). A reasonable simple choice for the matrices Q and R is given by the *Bryson's rule* [3, p. 537]: Select Q and R diagonal with

$$Q_{ii} = \frac{1}{\text{maximum acceptable value of } z_i^2}, \qquad i \in \{1, 2, \ldots, \ell\},$$
$$R_{jj} = \frac{1}{\text{maximum acceptable value of } u_j^2}, \qquad j \in \{1, 2, \ldots, k\},$$

which corresponds to the following criterion

$$J_{\mathrm{LQR}} := \int_0^\infty \Big(\sum_{i=1}^{\ell} Q_{ii}\, z_i(t)^2 + \rho \sum_{j=1}^{m} R_{jj}\, u(t)^2 \Big) dt.$$

In essence, the Bryson's rule scales the variables that appear in J_{LQR} so that the *maximum acceptable value for each term is one.* This is especially important when the units used for the different components of u and z make the values for these variables numerically very different from each other.

Although Bryson's rule usually gives good results, often it is just the starting point to a trial-and-error iterative design procedure aimed at obtaining desirable properties for the closed-loop system. We shall pursue this in Section 3.3. □

Additionally I would love to see if I could find the best values from some sort of optimizer code. I want to see if I can cook up something to find and compute K gain and a resulting stability score from each trial using a random or grid search algorithm. I would want to implement both Bryson's Rule and this optimization because Bryson's rule would give a great estimation based on real physical limits of the controller. Then from there, we could explore the tradeoffs around this baseline solution from a grid search algorithm. The workflow for this implementation would be to choose allowable limits based on the controllers allowable limits, compute the Bryson weights, and run a constrained parameter search to maximize stability and minimize unnecessary cart motion.